GMI German-Malaysian Institute

DIPLOMA PROGRAMME

PRACTICAL TEST

Academic Period : January 2025

Code	SPG 0453	Course Name	SECURED OBJECT ORIENTED PROGRAMMING
Title	PRACTICAL TEST	Examiner	MS HAZNI YATI MOHD GHADZI
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No	Assessment Criteria	CLO	PO	Level of Diff	Full Marks	Score	Marks	Comment
1	Ability to apply looping statement	2	2	P4				
	Ability to run with correct output							
2	Ability to read input	2	4	P4				
	Ability to apply selection statement							
	Ability to apply looping statement							
	Ability to apply looping for exit function							
	Ability to run with correct output							
3	Ability to read input	2	4	P4				
	Ability to apply selection statement for menu							
	Ability to apply looping for exit function							
	Ability to run with correct output							
				Total	50			

	Total Marks	
Total	PO 4	/50
	Signature	

Question 1 (10 Marks)

Write a Java program to display the alphabet Z to A using looping statement.

Question 2 (20 Marks)

Develop a Java application that is able to count the number of even and odd integer inserted by a user. User should be able to insert the number of integer he/she needs and insert the list of integers based on the inserted number. **Display the number of even and odd integers** counted by the program. Add in feature to repeat or exit the program.

Question 3 (20 Marks)

Develop a Java application that is able to **calculate the price of an item after SST (6%) and price after discount.** User must be able to **select the discount rate based on a menu given**. Display the new price and all necessary output. Add in the feature to repeat or exit the program. Hint input: Item name, item price, item quantity, discount rate.

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RUBRIC QUESTION 1					
CRITERIA	**MARKS (M)**				**MARKS ALLOCATED**
	0	**1-2**	**3-4**	**5**	
Ability to apply looping statement	Unable to apply looping condition	Incorrect application of looping statement	Correct implementation of looping statement with logical error	Able to apply looping condition	5
Ability to run with correct output	Program does not execute	Program executes with logic error	Program executes without logic error and display incorrect output	Program executes correctly without any logic error and display appropriate output	5
TOTAL					**/10**

RUBRIC QUESTION 2					
CRITERIA	**MARKS (M)**				**MARKS ALLOCATED**
	0	**1**	**2-3**	**4**	
Ability to read input	No input applied to the program	Unable to read the input	Able to read the input but with minimal inputs	Able to read the input with sufficient inputs	4
Ability to apply selection statement if else	Unable to apply if else statement	Unable to apply if else statement correctly	Able to apply required if else statement but produce incorrect results	Able to apply required if else statement and produce correct results	4
Ability to apply looping statement	Unable to apply looping statement	Unable to apply looping statement in a correct syntax	Able to apply looping statement but produce incorrect results	Able to apply looping statement and produce correct results	2
Ability to apply looping for exit function	Unable to apply looping for exit function	Unable to apply looping statement in a correct syntax	Able to apply looping but exit function failed	Able to apply for exit function	4
Ability to run with correct output	Program does not execute	Program executes with logic error	Program executes without logic error and display correct output partially	Program executes correctly without any logic error and display appropriate & correct output	1
TOTAL					**/20**

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RUBRIC QUESTION 3					
CRITERIA	**MARKS (M)**				**MARKS ALLOCATED**
	0	**1-2**	**3-4**	**5**	
Ability to read input	No input applied to the program	Unable to read the input	Able to read the input but with minimal inputs	Able to read the input with sufficient inputs	5
Ability to apply selection statement for menu	Unable to apply selection statement	Unable to apply selection statement correctly	Able to apply required selection structure but produce incorrect results	Able to apply required selection structure and produce correct results	5
Ability to apply looping for exit function	Unable to apply looping for exit function	Unable to apply looping statement in a correct syntax	Able to apply looping but exit function failed	Able to apply for exit function	5
Ability to run with correct output	Program does not execute	Program executes with logic error	Program executes without logic error and display correct output partially	Program executes correctly without any logic error and display appropriate & correct output	1
TOTAL					**/20**

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ANSWER FOR QUESTION 1 – SCREENSHOT OF CODING & OUTPUT

Source Code | Question 1

```
package com.mycompany.pt1;

/**
 *
 * @author abdul
 */

import javax.swing.*;
import java.util.Scanner;

public class PT1 {

    public static void main(String[] args) {

        for (char letter = 'Z'; letter >= 'A'; letter--){
        System.out.print(" " + letter);

        }
    }
}
```

Output | Question 1

```
--- exec:3.1.0:exec (default-cli) @ PT1 ---
 Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
------------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------------
Total time:  1.672 s
Finished at: 2025-03-27T10:04:46+08:00
------------------------------------------------------------------------
```


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ANSWER FOR QUESTION 2 – SCREENSHOT OD CODING & OUTPUT

SOURCE CODE

```
public class PT2 {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);
        int loop;

        int evenCount = 0;
        int oddCount = 0;

        do { //Lopp start sini


            //User input
            System.out.print("Enter the number of integers you want to insert: ");
            int numcount = scanner.nextInt();

               //for loop
            System.out.println("Enter " + numcount + " numbers:");
            for (int i = 0; i < numcount; i++) {
                System.out.print("Number " + (i + 1) + ": ");
                int number = scanner.nextInt();

                    //Sini dia check utk event dgn odd
                if (number % 2 == 0) {
                    evenCount++;
                } else {
                    oddCount++;
                }
            }


            System.out.println("\nResults:");
            System.out.println("Even numbers count: " + evenCount);
            System.out.println("Odd numbers count: " + oddCount);
            System.out.print("\nDo you want to repeat? (1 = Yes, 0 = No): ");
            loop = scanner.nextInt();

        } while (loop == 1);

        System.out.println("Exiting Program..");
        scanner.close();
    }
}
```

OUTPUT :

```
--- exec:3.1.0:exec (default-cli) @ PT2 ---
Enter the number of integers you want to insert: 3
Enter 3 numbers:
Number 1: 3
Number 2: 4
Number 3: 5

Results:
Even numbers count: 1
Odd numbers count: 2
```


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```
Do you want to repeat? (1 = Yes, 0 = No): 0
Exiting Program..
------------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------------
Total time:  23.397 s
Finished at: 2025-03-27T10:18:57+08:00
------------------------------------------------------------------------
```


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ANSWER FOR QUESTION 3 – SCREENSHOT OF CODING & OUTPUT

SOURCE CODE | QUESTION 3

```
import javax.swing.*;
[redacted]

public class PTJ {

    public static void main(String[] args) {

        int choice = JOptionPane.showConfirmDialog(null, "Welcome to Discount Calculator!\n\nProceed ?", "Discount Calculator", JOptionPane.YES_NO_OPTION, JOptionPane.INFORMATION_MESSAGE);

        if (choice == JOptionPane.YES_OPTION) {
            Input();
        }else{
            Exit();
        }

    }

    public static void Input() {
        int exit;

        JOptionPane.showMessageDialog(null, "You are required to enter the following:\n\n1. Item Name\n2. Price\n3. Quantity\n4. Discount Rate");

        do { //Loop utk  menu

            String item_name = JOptionPane.showInputDialog(null, "Enter Item Name:", "Item Name", JOptionPane.QUESTION_MESSAGE);
            double item_price;
                    item_price = Double.parseDouble(JOptionPane.showInputDialog(null, "Enter Price:", "Price", JOptionPane.QUESTION_MESSAGE));

            int item_quantity;
                    String item_quantity_input = JOptionPane.showInputDialog(null, "Enter Quantity:", "Quantity", JOptionPane.QUESTION_MESSAGE);
                    item_quantity = Integer.parseInt(item_quantity_input);

            String[] options = {"15%", "20%"};
            int discount_choice = JOptionPane.showOptionDialog(null, "Please choose a discount rate:", "Discount Rate",
                    JOptionPane.DEFAULT_OPTION, JOptionPane.QUESTION_MESSAGE, null, options, options[0]); //Menu untuk tekan butang discount

            double discount_rate = (discount_choice == 0) ? 0.15 : 0.20; //Pilih discount mana
            double initial_price = item_price * item_quantity;
            double discounted_price = initial_price * discount_rate;
            double final_price = initial_price - discounted_price;

            //Compile semua variable dlm satu String - receipt (kalau tak paning :P)
            String receipt = String.format(
                    "Here is your receipt:\n" +
                            "Item Name: %s\n" +
                            "Item Price: %.2f\n" +
                            "Quantity: %d\n" +
                            "Discount Rate: %.0f%%\n" +
                            "Final Price: %.2f",
                    item_name, item_price, item_quantity, discount_rate * 100, final_price
            );
            //Output
            JOptionPane.showMessageDialog(null, receipt, "Receipt", JOptionPane.INFORMATION_MESSAGE);

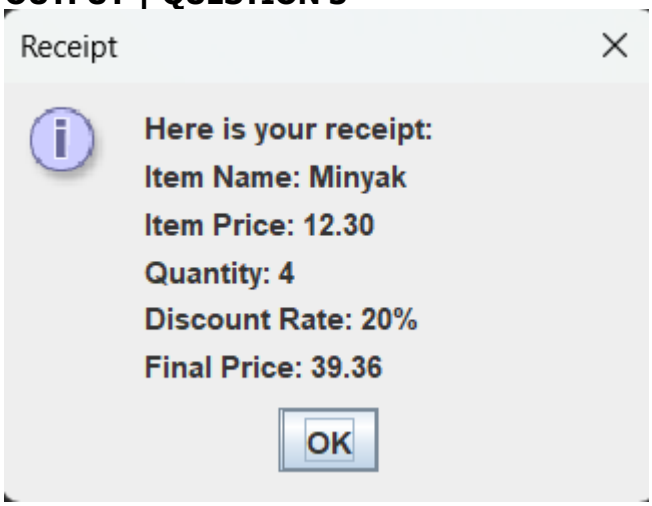
            //Confirmation utk repeat
            exit = JOptionPane.showConfirmDialog(null, "Would you like to enter another item?", "Confirmation", JOptionPane.YES_NO_OPTION, JOptionPane.QUESTION_MESSAGE);

        } while (exit == JOptionPane.YES_OPTION);
    }



    private static void Exit() {
        return;
    }
}
```

OUTPUT | QUESTION 3



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